

NORTH BATTLEFORD, SK, Canada

WMO: 718760

Lat: 52.7691N

Lon: 108.2510W

Elev: 1799

StdP: 13.77

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.8	-22.7	-36.2	0.7	-27.5	-30.7	1.0	-22.4	28.7	12.8	25.3	13.8	4.7	300	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	84.8	64.4	81.3	63.0	78.3	61.9	67.8	79.6	65.7	77.0	63.8	74.6	11.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	94.1	73.4	61.6	87.4	71.0	59.5	81.3	68.9	33.3	79.9	31.6	77.2	30.0	74.5	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	22.4	20.1	DB	-34.2	92.0	5.8	4.5	-38.4	95.2	-41.8	97.8	-45.1	100.4	-49.3	103.6	(4)
(5)			WB	-34.4	71.9	5.7	2.4	-38.6	73.6	-41.9	75.0	-45.1	76.3	-49.3	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	5.1	8.2	20.2	38.2	51.2	59.6	64.1	62.1	53.0	38.9	21.8	8.6		
	DBStd	24.00	15.88	13.89	14.04	10.52	8.28	6.11	5.29	6.23	8.19	8.88	12.13	14.56		
	HDD50	6499	1392	1171	923	371	85	5	0	2	60	358	848	1284		
	HDD65	10683	1857	1591	1387	803	432	181	80	130	366	810	1297	1749		
	CDD50	1411	0	0	0	17	123	292	438	377	151	13	0	0		
	CDD65	123	0	0	0	0	5	18	53	41	6	0	0	0		
	CDH74	1825	0	0	0	9	132	288	634	581	177	4	0	0		
	CDH80	446	0	0	0	1	25	61	150	165	44	0	0	0		
	WSAvg	9.1	8.4	8.4	9.9	10.7	10.4	9.3	8.3	8.2	8.9	9.3	8.8	8.2		
	PrecAvg	15.4	0.8	0.6	0.7	0.9	1.5	2.6	2.7	2.0	1.3	0.6	0.7	0.8		
(15) Precipitation	PrecMax	19.5	1.6	1.5	1.5	3.2	5.4	7.3	5.7	5.0	3.3	1.8	1.7	1.5		
	PrecMin	11.3	0.2	0.1	0.2	0.0	0.3	0.3	0.5	0.3	0.1	0.1	0.1	0.2		
	PrecStd	2.1	0.4	0.4	0.4	0.7	0.9	1.6	1.2	1.5	0.9	0.4	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	40.5	54.4	73.4	83.3	85.9	88.6	90.5	85.2	72.1	53.5	43.1		
		MCWB	35.8	35.1	43.5	52.5	58.4	63.4	68.0	65.0	61.9	54.4	44.0	37.3		
	2%	DB	37.2	35.9	45.8	67.2	77.7	80.8	84.0	84.5	79.2	63.6	46.4	38.0		
		MCWB	33.3	32.1	38.7	49.8	56.3	61.4	66.5	64.5	59.3	50.0	39.6	33.5		
	5%	DB	32.8	31.7	41.4	61.3	73.0	77.0	80.5	80.1	73.9	58.0	41.3	33.2		
		MCWB	30.0	29.3	36.1	46.4	54.3	59.5	65.0	63.6	57.2	47.0	36.5	30.2		
	10%	DB	27.3	27.9	37.6	55.7	68.6	73.5	77.3	76.6	68.8	53.0	37.1	28.5		
		MCWB	25.5	26.0	33.9	43.5	52.2	58.2	63.8	61.8	54.8	44.7	33.7	26.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.2	35.5	44.0	54.4	61.7	67.3	72.0	69.3	63.7	55.7	44.3	37.7		
		MCDB	40.3	40.1	52.7	71.2	77.9	79.3	83.7	81.5	79.3	69.5	51.8	42.1		
	2%	WB	33.4	32.3	39.2	50.5	57.7	64.1	68.9	66.9	60.4	51.2	40.4	33.8		
		MCDB	37.0	35.3	45.1	64.6	73.5	75.5	80.4	79.8	76.7	60.5	45.8	37.5		
	5%	WB	29.9	29.3	36.3	47.4	55.4	61.9	66.8	64.9	58.2	48.0	36.8	30.1		
		MCDB	32.8	31.6	40.8	59.9	70.5	72.9	77.4	77.1	71.8	56.7	40.6	33.2		
	10%	WB	25.5	26.0	34.0	44.5	53.1	60.0	64.8	62.9	55.7	45.2	33.7	26.3		
		MCDB	27.3	27.9	37.5	54.6	66.3	70.7	74.9	74.3	66.9	52.3	36.9	28.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.8	17.5	21.4	24.6	21.8	22.1	23.6	23.6	19.9	15.8	15.9		
		MCDBR	19.9	20.7	20.7	30.9	31.5	28.7	27.8	30.4	33.1	29.2	21.5	20.8		
		MCWBR	16.8	18.0	14.9	17.0	15.3	13.4	13.8	14.2	16.3	17.5	15.7	17.5		
		MCDBR	19.5	19.8	19.8	29.4	27.9	23.6	24.4	26.3	30.5	26.8	20.2	19.6		
	5% WB	MCWBR	16.8	17.5	14.6	16.9	14.1	12.4	13.3	13.3	15.6	16.8	15.2	16.6		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.252	0.291	0.341	0.360	0.375	0.374	0.378	0.318	0.279	0.255	0.224		
	taud		2.266	2.332	2.362	2.338	2.335	2.331	2.347	2.323	2.472	2.553	2.452	2.364		
	Ebn at Noon		235	279	287	283	282	278	276	268	275	268	242	236		
	Edh at Noon		19	25	30	35	37	38	37	36	27	20	16	15		
(44) All-Sky Solar Radiation	RadAvg		334	671	1144	1506	1848	1883	1966	1597	1140	668	328	242		
	RadStd		32	43	89	144	146	144	114	130	121	104	41	26		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page